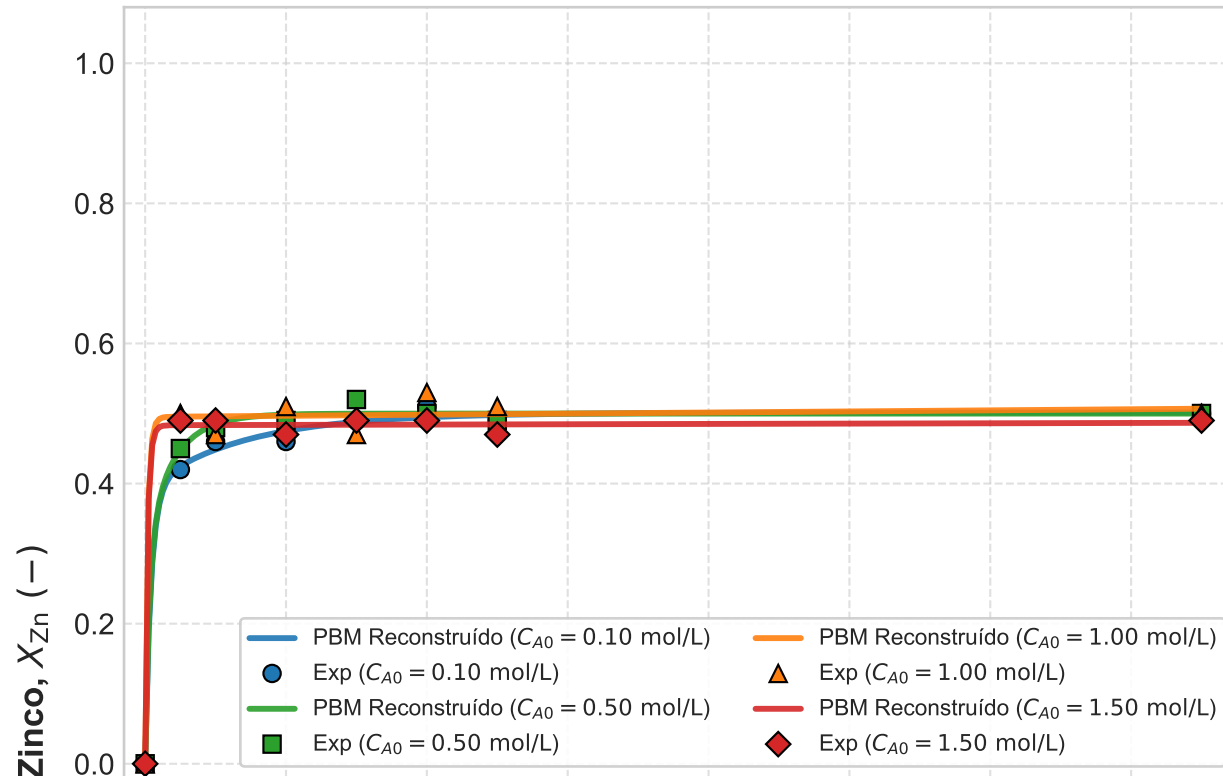


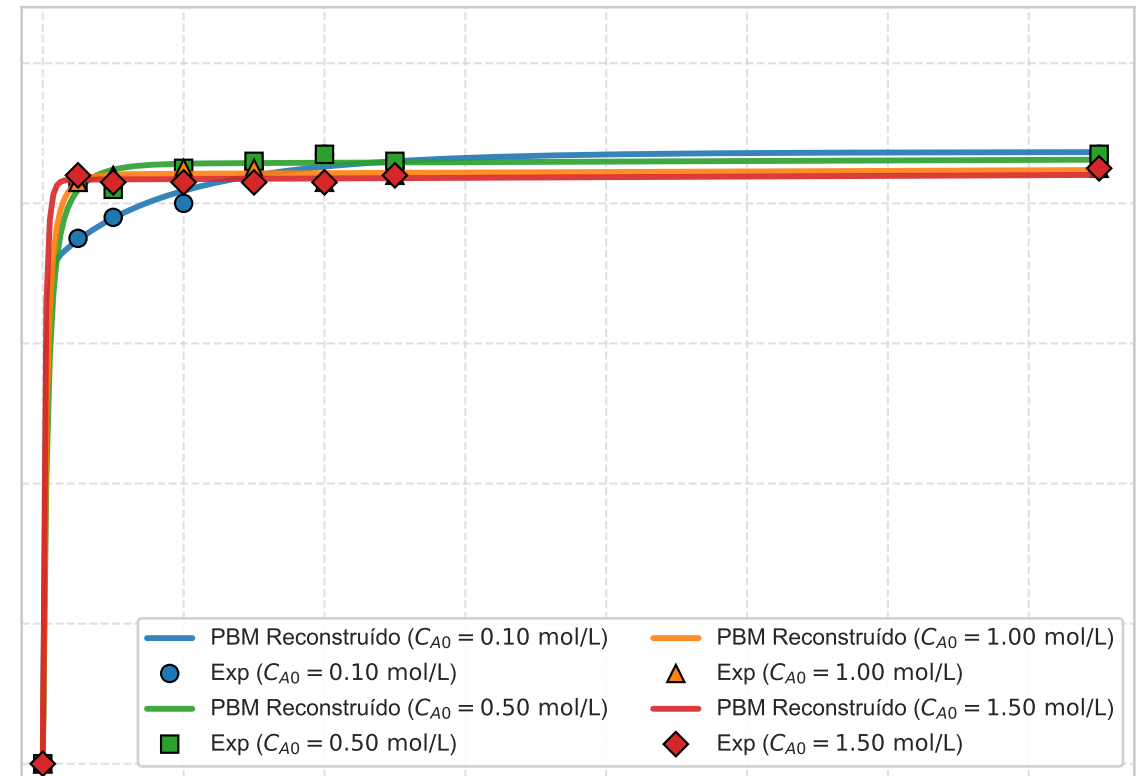
Validação da Inversão: Conversão Reconstituída via PBM com $v(t)$ Otimizado vs. Dados Experimentais

Aderência Global de Reconstrução: $R^2 = 0.99896$, RMSE = 0.0105, MAE = 0.0071

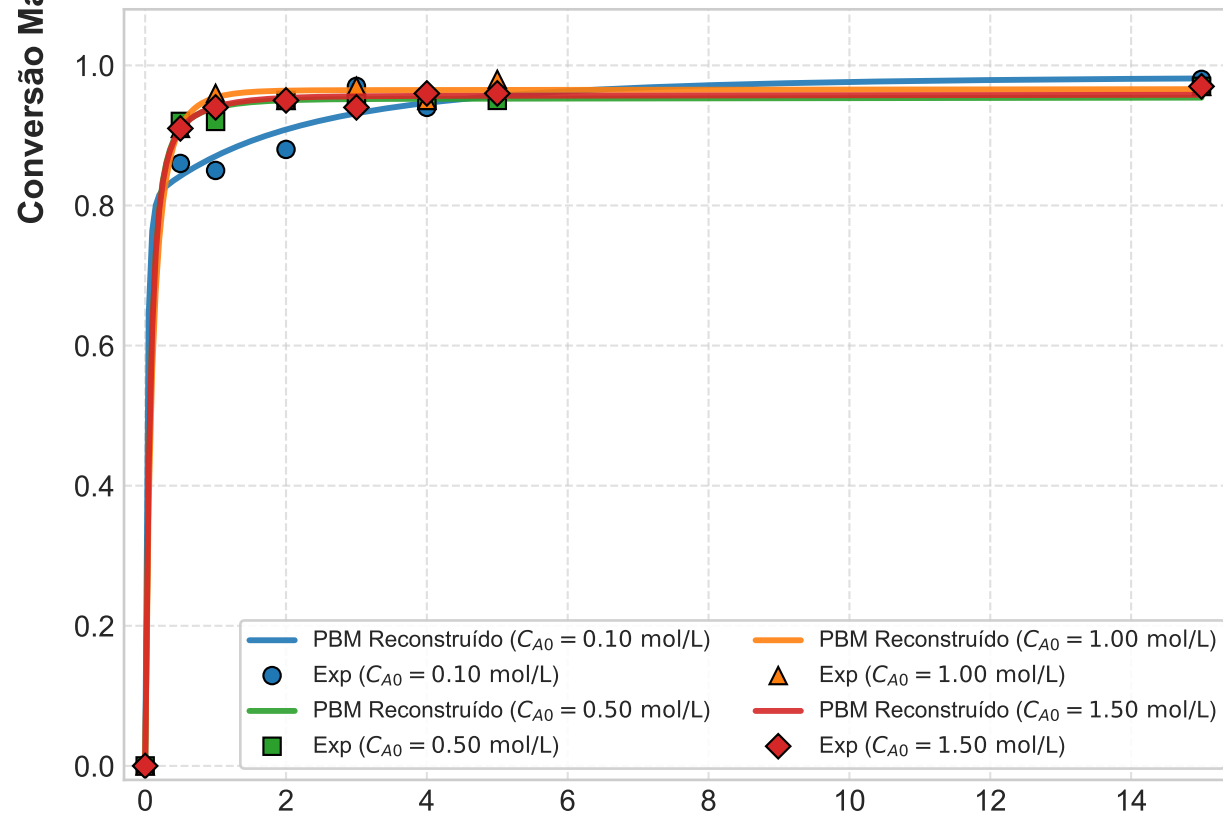
Painel (a) — Razão Molar $\eta = 0.5$



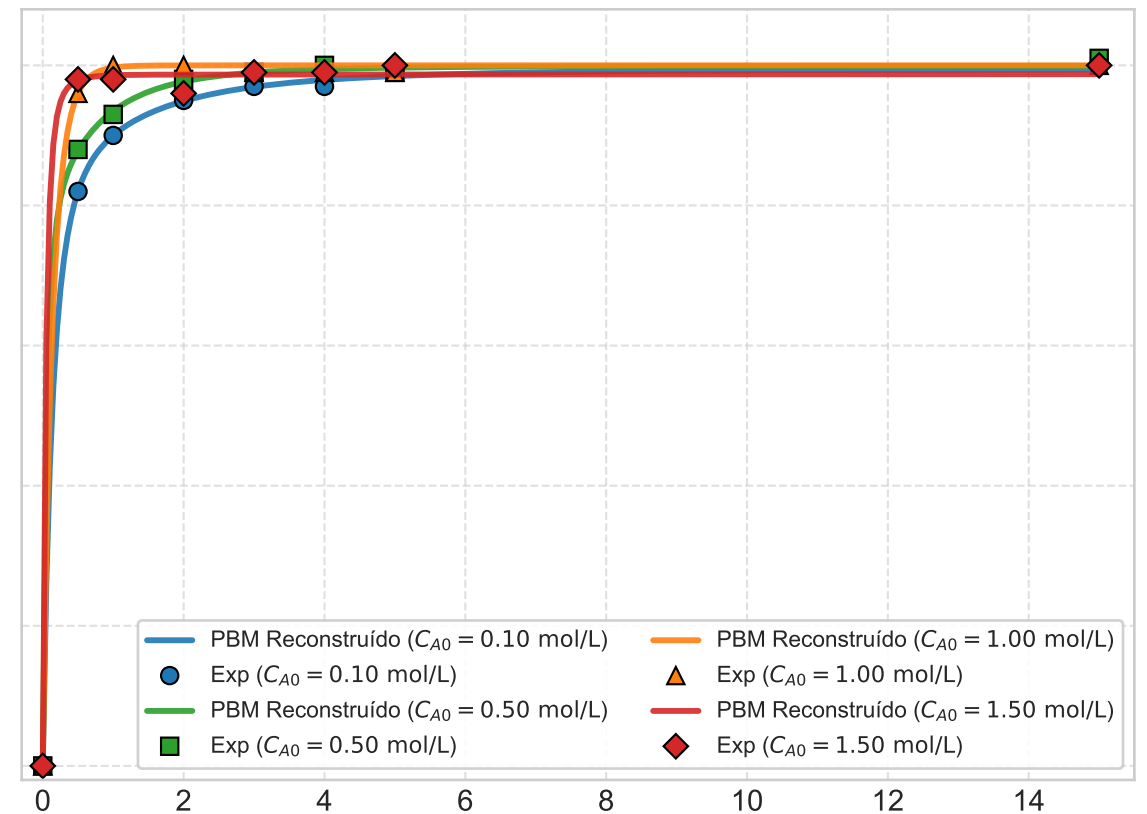
Painel (b) — Razão Molar $\eta = 1.0$



Painel (c) — Razão Molar $\eta = 1.5$



Painel (d) — Razão Molar $\eta = 3.1$



Tempo de Reação, t (min)